

Assignment 1.4: Pythagorean Thm

1. Find the square root of each number below.

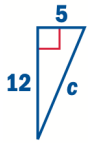
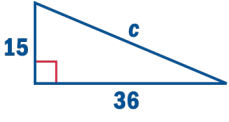
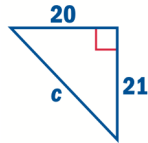
$\sqrt{49} = \underline{\hspace{1cm}}$

$\sqrt{121} = \underline{\hspace{1cm}}$

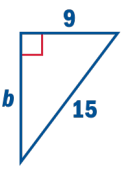
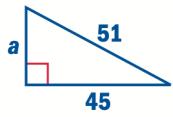
$\sqrt{900} = \underline{\hspace{1cm}}$

$\sqrt{6.25} = \underline{\hspace{1cm}}$

2. Find the length of the hypotenuse in each right triangle below. Show all work.

 $c = \underline{\hspace{1cm}}$	 $c = \underline{\hspace{1cm}}$	 $c = \underline{\hspace{1cm}}$
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3. Find the length of the missing leg in each right triangle below. Show all work.

 $a = \underline{\hspace{1cm}}$	 $b = \underline{\hspace{1cm}}$	 $a = \underline{\hspace{1cm}}$
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4. In triangle DEF , angle F is a right angle. If $DE = 26$ and $EF = 24$, find the length of DF . Show all work.

5. **Application** – An 18-foot ladder leans against the wall of a building. The base of the ladder is 9 feet from the building on level ground. How many feet up the wall, to the nearest tenth of a foot, is the top of the ladder? Show all work.

6. **Exploration** – If the three sides of a triangle satisfy the equation $a^2 + b^2 = c^2$, then the triangle is a *right triangle*. Remember, the longest side must be the hypotenuse!

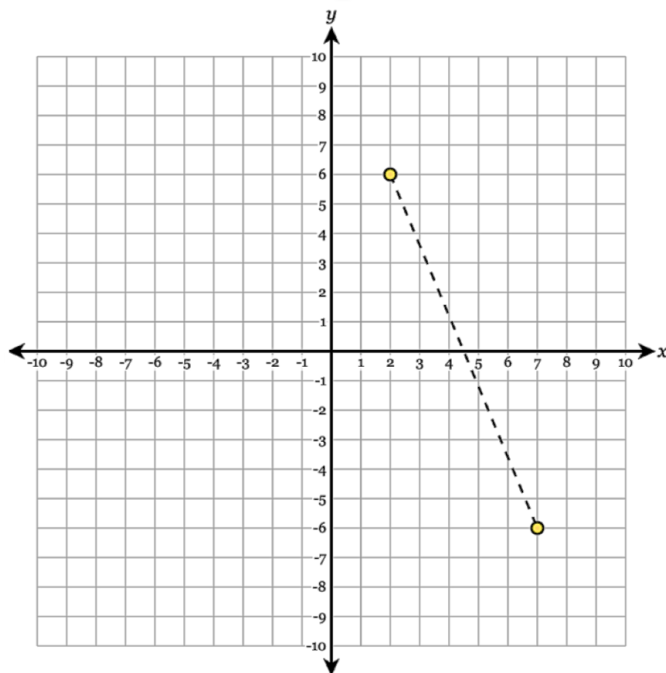
Determine whether a triangle with given side lengths is a right triangle. Show all work.

$$a = 16, b = 12, c = 20$$

$$a = 4, b = 5, c = 6$$

$$a = 15, b = 15, c = 30$$

7. Determine the distance between the points shown in the graph below. Round to the nearest tenth if necessary.



Optional Challenge

8. Investigation – A Pythagorean Triple is a set of 3 positive integers (no decimals) that satisfy Pythagorean's Theorem. You can create a Pythagorean Triple by substituting any integer *greater than* 1 into each of the following equations: $a = 2n$, $b = n^2 - 1$, $c = n^2 + 1$.

a) Choose any integer n greater than 2 and substitute your value into each expression.

b) Verify (check) that your answers satisfy Pythagorean's Theorem.

9. Analysis – Your friend claims that he has found a way to use an existing Pythagorean Triple to find a new one. All you need to do is multiply the length of each leg by 2 and multiply the length of the hypotenuse by 4. Do you agree or disagree with your friend? Justify your answer.